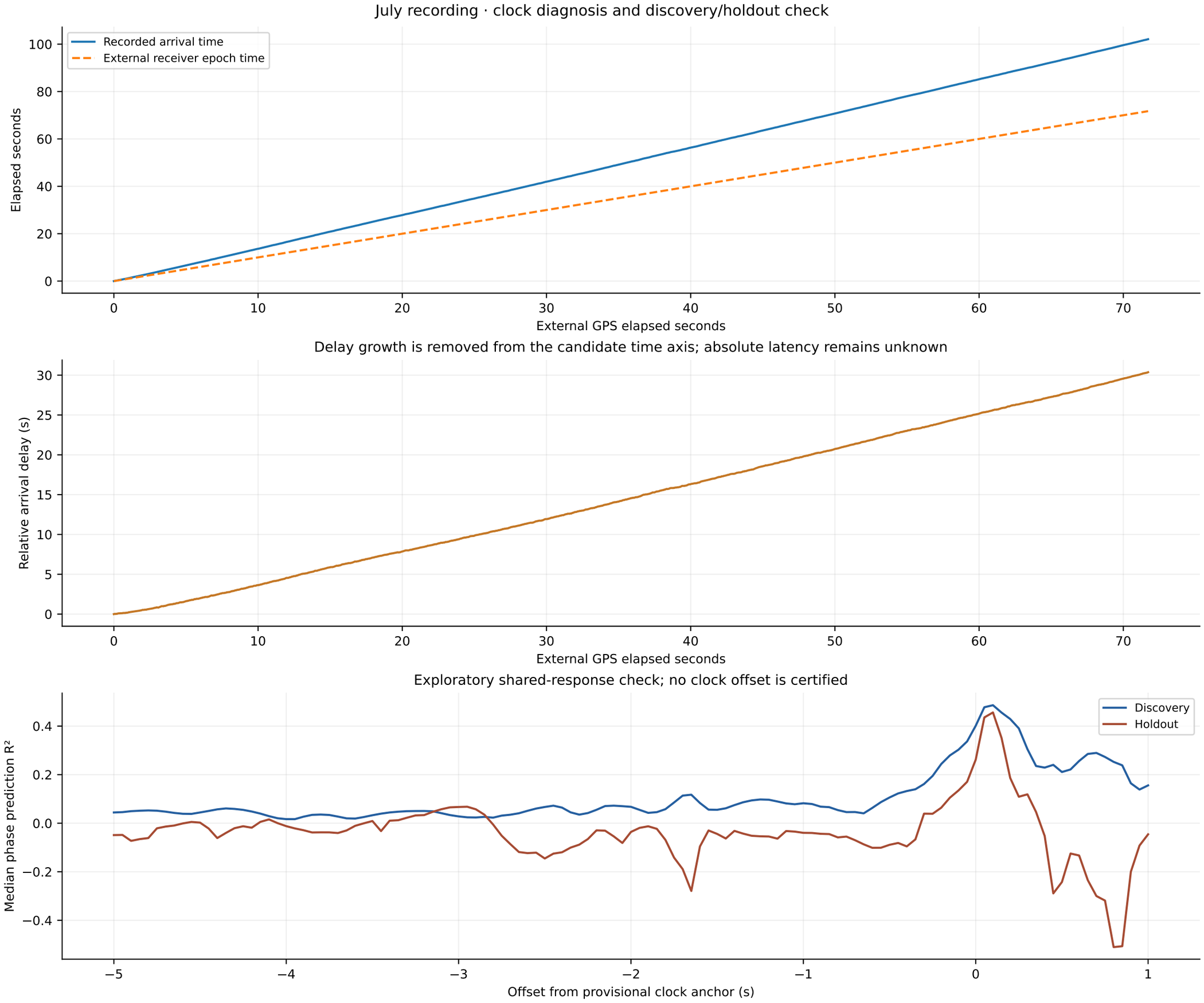
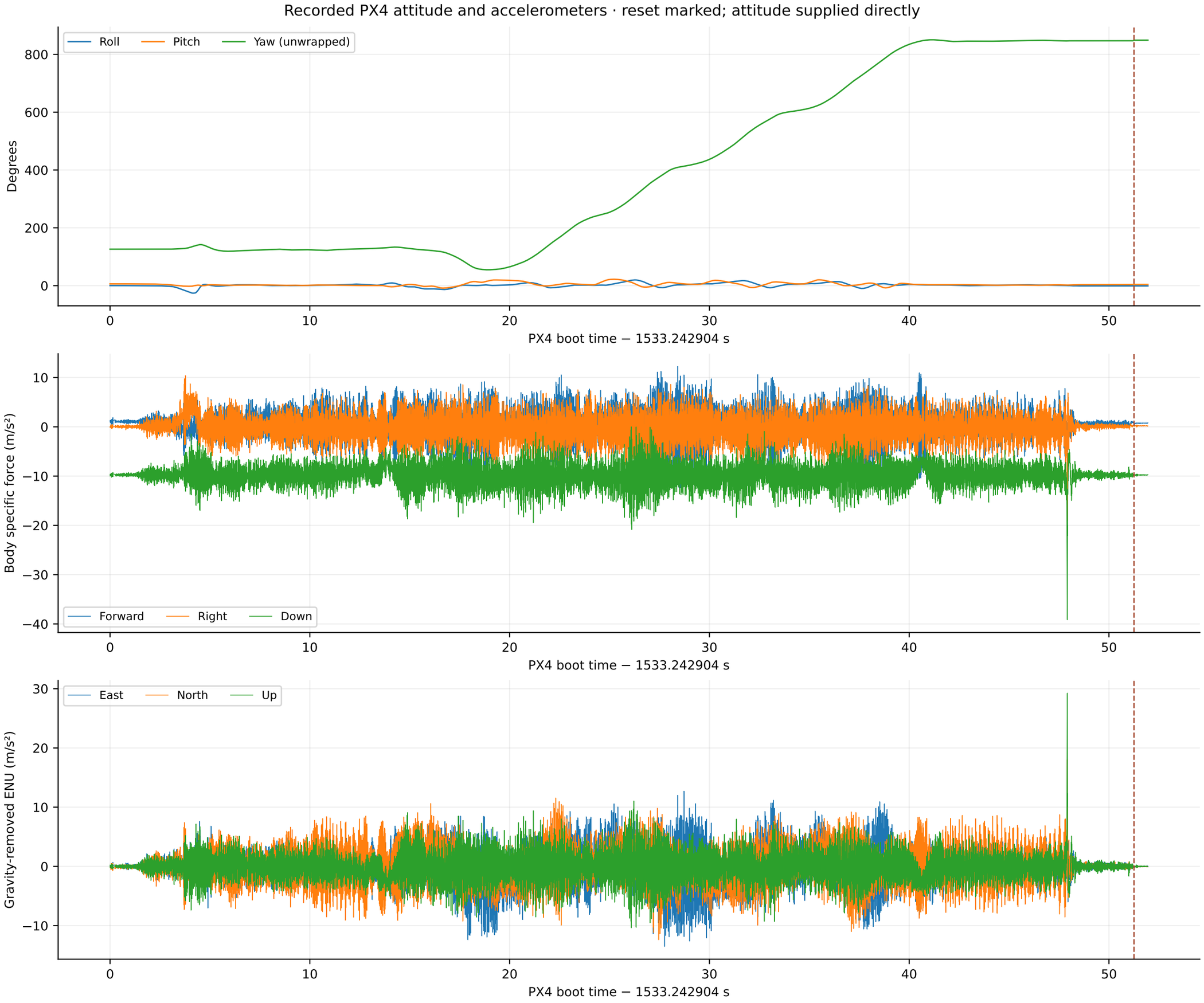
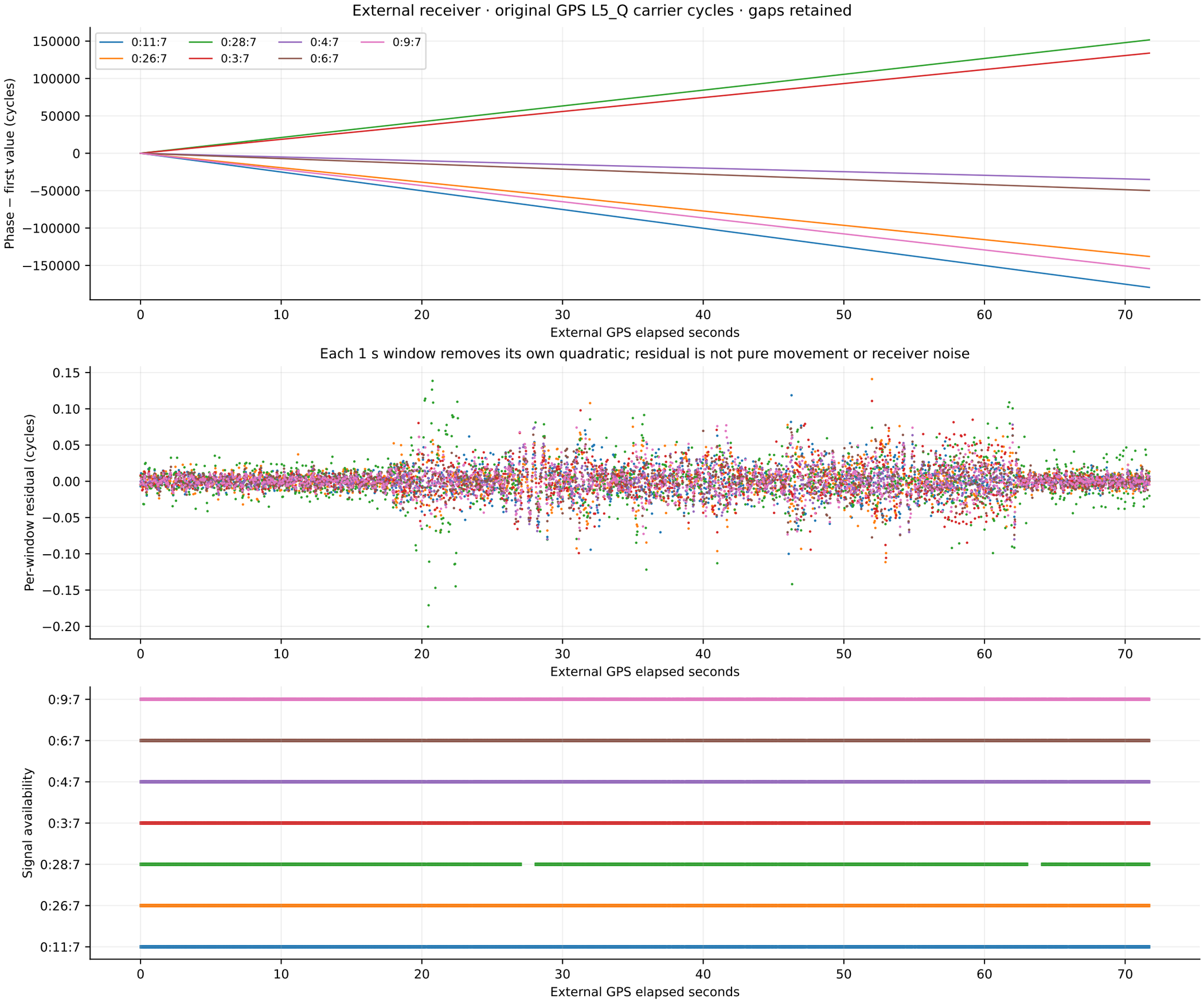
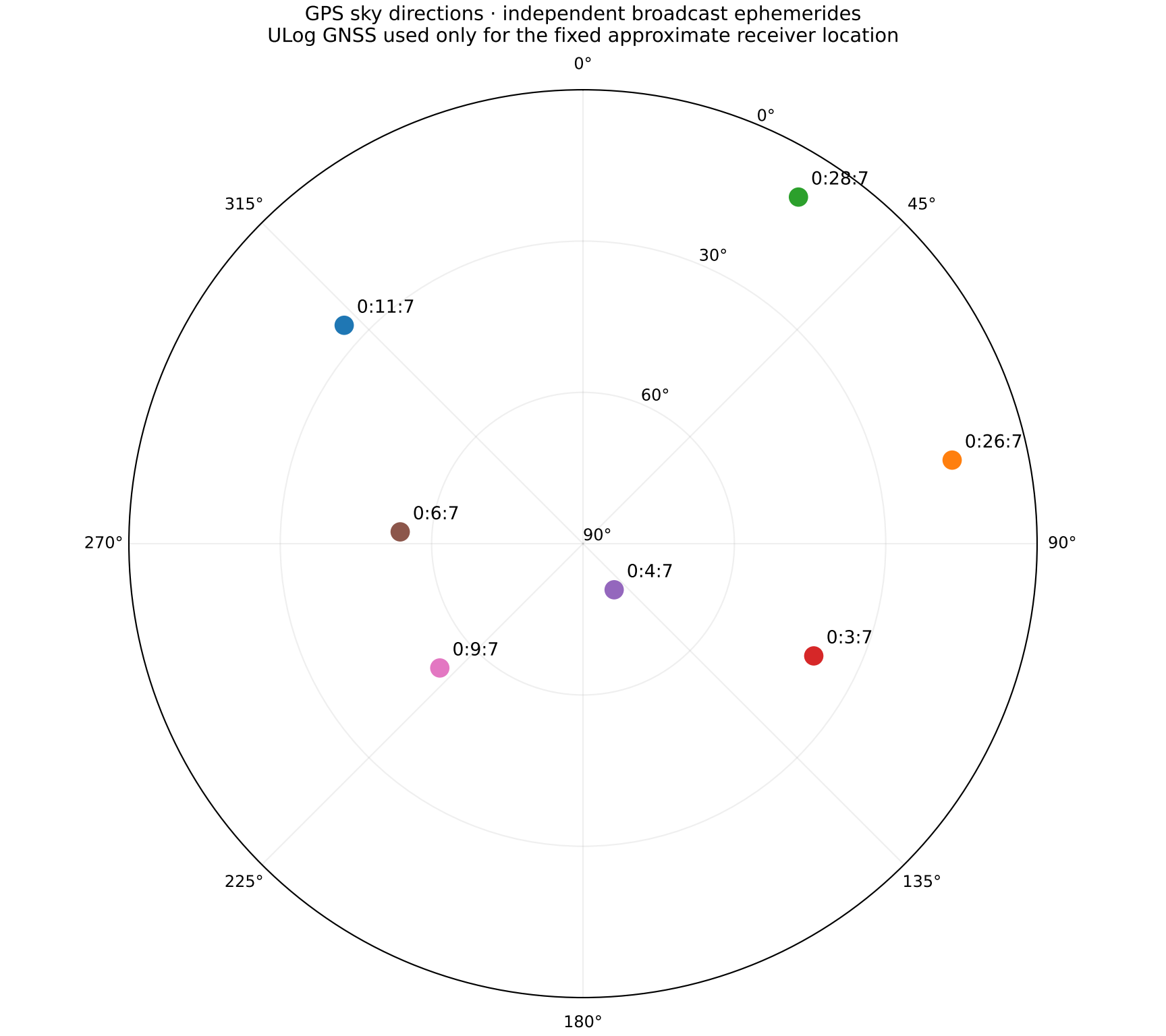
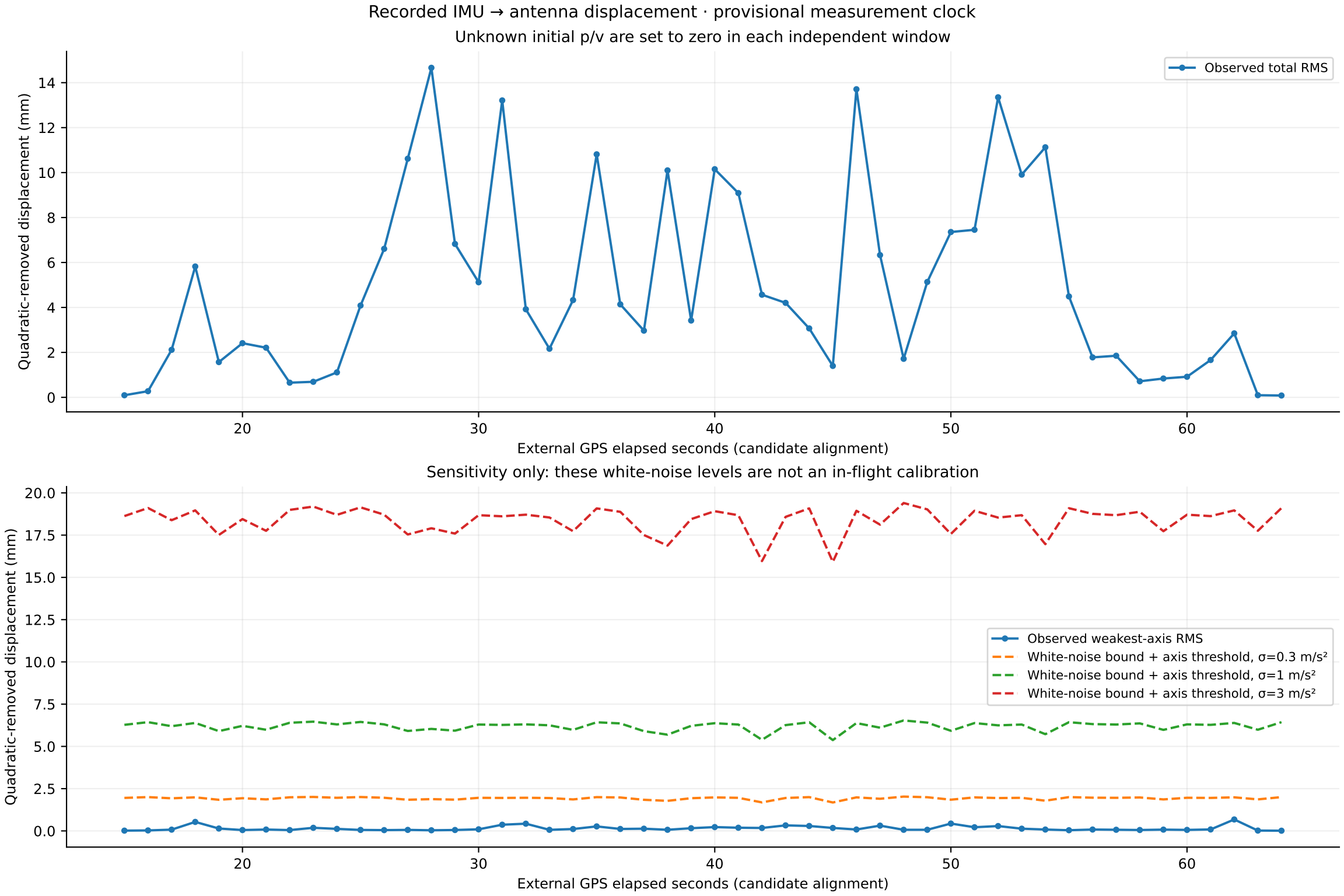
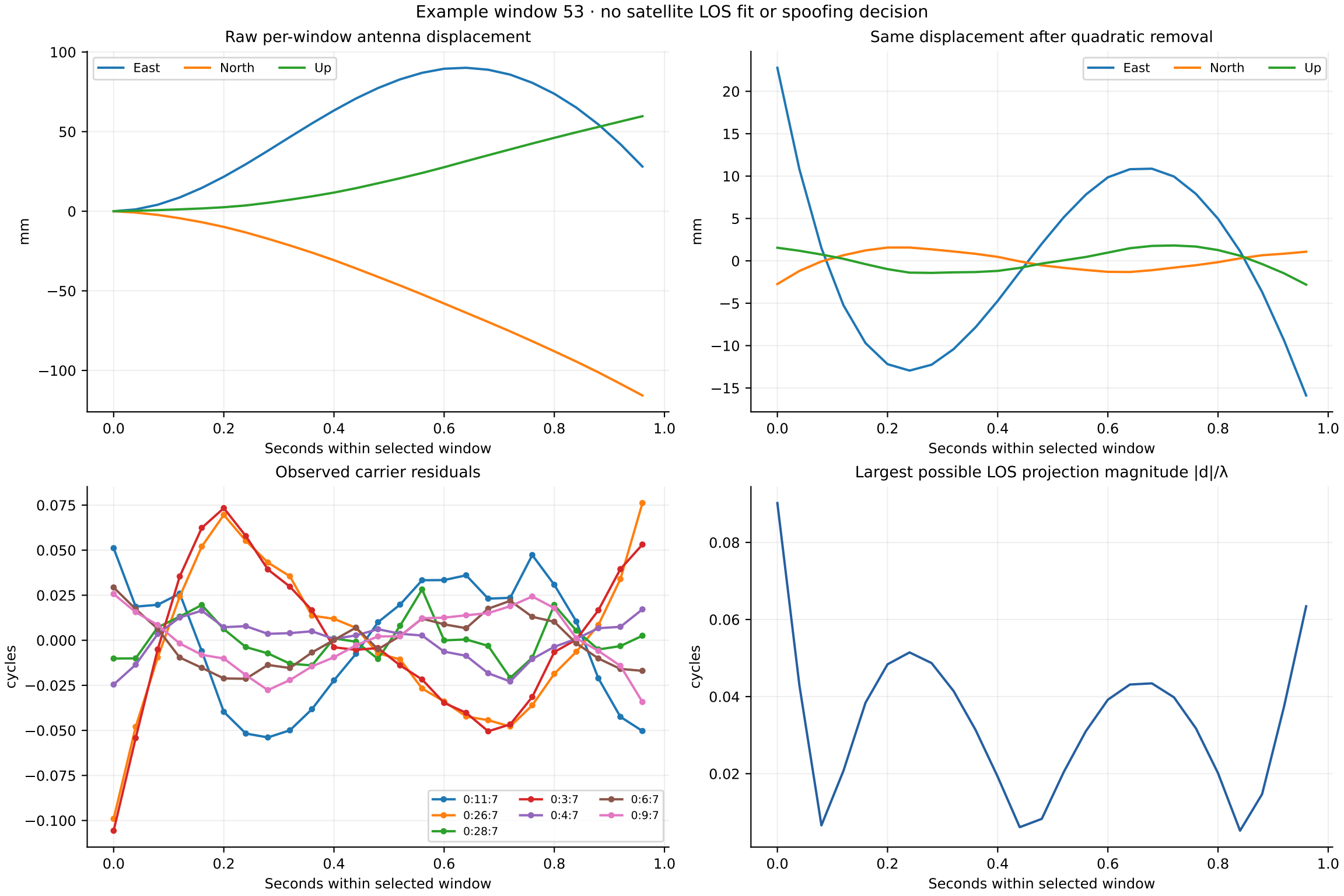

July recording · clock diagnosis and discovery/holdout check
Recorded arrival time
External receiver epoch time
Elapsed seconds
External GPS elapsed seconds
Delay growth is removed from the candidate time axis; absolute latency remains unknown
Relative arrival delay (s)
External GPS elapsed seconds
Exploratory shared-response check; no clock offset is certified
Discovery
Holdout
Median phase prediction R²
Offset from provisional clock anchor (s)


Recorded PX4 attitude and accelerometers · reset marked; attitude supplied directly
Roll
Pitch
Yaw (unwrapped)
Degrees
800
600
400
200
0
0
10
20
30
40
50
PX4 boot time − 1533.242904 s
Body specific force (m/s²)
10
0
−10
−20
−30
−40
Forward
Right
Down
0
10
20
30
40
50
PX4 boot time − 1533.242904 s
Gravity-removed ENU (m/s²)
East
North
Up
30
20
10
0
−10
0
10
20
30
40
50
PX4 boot time − 1533.242904 s


External receiver · original GPS L5_Q carrier cycles · gaps retained
0:11:7
0:26:7
0:28:7
0:3:7
0:4:7
0:6:7
0:9:7
Phase − first value (cycles)
150000
100000
50000
0
−50000
−100000
−150000
0
10
20
30
40
50
60
70
External GPS elapsed seconds
Each 1 s window removes its own quadratic; residual is not pure movement or receiver noise
Per-window residual (cycles)
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
External GPS elapsed seconds
Signal availability
0:9:7
0:6:7
0:4:7
0:3:7
0:28:7
0:26:7
0:11:7
External GPS elapsed seconds


GPS sky directions · independent broadcast ephemerides
ULog GNSS used only for the fixed approximate receiver location
0°
0°
45°
90°
135°
180°
225°
270°
315°
30°
60°
90°
0:28:7
0:11:7
0:26:7
0:6:7
0:4:7
0:3:7
0:9:7


Recorded IMU → antenna displacement · provisional measurement clock
Unknown initial p/v are set to zero in each independent window
Observed total RMS
Quadratic-removed displacement (mm)
External GPS elapsed seconds (candidate alignment)
Sensitivity only: these white-noise levels are not an in-flight calibration
Observed weakest-axis RMS
White-noise bound + axis threshold, σ=0.3 m/s²
White-noise bound + axis threshold, σ=1 m/s²
White-noise bound + axis threshold, σ=3 m/s²
Quadratic-removed displacement (mm)
External GPS elapsed seconds (candidate alignment)


Example window 53 · no satellite LOS fit or spoofing decision
Raw per-window antenna displacement
East
North
Up
mm
Seconds within selected window
Same displacement after quadratic removal
East
North
Up
mm
Seconds within selected window
Observed carrier residuals
cycles
0:11:7
0:26:7
0:28:7
0:3:7
0:4:7
0:6:7
0:9:7
Seconds within selected window
Largest possible LOS projection magnitude |d|/λ
cycles
Seconds within selected window